

0/1 Knapsack - A Visual Walkthrough

We have a knapsack with capacity $W = 8$ and 4 items. Each item has a weight and a value. We can either take an item (1) or leave it (0). Goal: maximize total value without exceeding capacity.

Our Items

Item 1	Item 2	Item 3	Item 4
$w=2$ $v=3$	$w=3$ $v=4$	$w=4$ $v=5$	$w=5$ $v=6$

The Empty Sack

We start with an empty sack. Capacity = 8, current weight = 0, current value = 0.



Sack: $W=$
Current
Current

The DP Table

We build a table $dp[i][j] = \text{max value using first } i \text{ items with capacity } j$.

Rows = items (0 to 4), Columns = capacity (0 to 8).

$$dp[i][j] = \max(dp[i-1][j], dp[i-1][j-w_i] + v_i)$$

exclude item i include item i

Step 1: Initialize - Row 0 and column 0 are all zeros

No items = 0 value. Zero capacity = 0 value.

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0								
2	0								
3	0								
4	0								

Step 2: Consider Item 1 (w=2, v=3)

considering...

Item 1	Item 2	Item 3	Item 4
w=2 v=3	w=3 v=4	w=4 v=5	w=5 v=6

For each capacity j from 0 to 8:

If $j < 2$: can't include Item 1, so $dp[1][j] = dp[0][j] = 0$

If $j \geq 2$: $dp[1][j] = \max(dp[0][j], dp[0][j-2] + 3)$

Let's compute $dp[1][2]$:

Item 1 (w=2, v=3)
 $dp[0][0] + 3 = \text{Exclude} = dp[0][2] =$
 $\gg \text{INCLUDE Item 1} \ll$

Include wins! $dp[1][2] = 3$. Similarly, for all $j \geq 2$, we

can fit Item 1.

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	0	3	3	3	3	3	3	3
2	0								
3	0								
4	0								

Step 3: Consider Item 2 ($w=3, v=4$)

considering...

Item 1	Item 2	Item 3	Item 4
$w=2, v=3$	$w=3, v=4$	$w=4, v=5$	$w=5, v=6$

For $j < 3$: can't include Item 2, carry forward from row 1.
 For $j \geq 3$: $dp[2][j] = \max(dp[1][j], dp[1][j-3] + v)$

Let's compute $dp[2][5]$ (capacity = 5):

Item 1 Item 2 ($w=3, v=4$)

$$dp[1][2] + 4 = \text{Exclude} \quad dp[1][5] =$$

>> INCLUDE Item 2 <<

Include wins! $dp[2][5] = 7$. This means: with items {1,2} and capacity 5, we can get value 7 (take both items: $w=2+3=5, v=3+4=7$).

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	0	3	3	3	3	3	3	3
2	0	0	3	4	4	7	7	7	7
3	0								
4	0								

Step 4: Consider Item 3 (W=4, V=5)

	Item 1	Item 2	Item 3	Item 4
	W=2 V=3	W=3 V=4	W=4 V=5	W=5 V=6

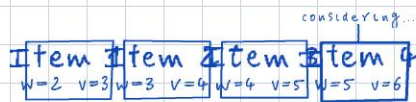
Key cell: $dp[3][6]$ (capacity = 6, considering items 1,2,3)

Item: Item 3 (W=4, V=5)
 $dp[2][2] + 5$ -> Excludes $dp[2][6]$ =
 >> INCLUDE Item 3 <<

Include wins! $dp[3][6] = 8$. We can get value 8 with capacity 6 using items {1,3}: $w=2+4=6$, $v=3+5=8$.

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	0	3	3	3	3	3	3	3
2	0	0	3	4	4	7	7	7	7
3	0	0	3	4	5	7	8	9	9
4	0								

Step 5: Consider Item 4 (W=5, V=6)



The final answer is $dp[4][8]$ - all items, full capacity.

Item: Item 4 (w=5, v=6)

$dp[3][3] + 6$ + Exclude: $dp[3][8] =$
 $\gg$ INCLUDE Item + $\ll$

Include wins! $dp[4][8] = 10$.

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	0	3	3	3	3	3	3	3
2	0	0	3	4	4	7	7	7	7
3	0	0	3	+	5	7	8	9	9
4	0	0	3	+	5	7	8	9	10

The Complete DP Table

Maximum value with all + items and capacity 8 is 10.

Backtracking: Which Items Did We Take?

Start at $dp[4][8] = 10$. Walk backwards to find which items were selected.

Rule: If $dp[i][j] \neq dp[i-1][j]$, then item i WAS taken.
 Jump to $dp[i-1][j - w_i]$ and continue.

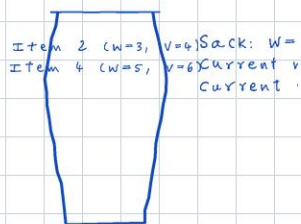
0	0	0	0	0	0	0	0	0
0	0	3	3	3	3	3	3	3
0	0	3	4	4	7	7	7	7
0	0	3	4	5	7	8	9	9
0	0	3	4	5	7	8	9	10

Trace:

- $dp[4][8]=10 \neq dp[3][8]=9 \rightarrow$ Item 4 WAS taken. Jump to $dp[3][8-5]=dp[3][3]=4$
- $dp[3][3]=4 == dp[2][3]=4 \rightarrow$ Item 3 was NOT taken. Stay at $dp[2][3]=4$
- $dp[2][3]=4 \neq dp[1][3]=3 \rightarrow$ Item 2 WAS taken. Jump to $dp[1][3-3]=dp[1][0]=0$
- $dp[1][0]=0 == dp[0][0]=0 \rightarrow$ Item 1 was NOT taken (at capacity 0)

Selected items: Item 2 and Item 4

The Final Sack



Weight: $3 + 5 = 8$ (exactly fills capacity!)

Value: $4 + 6 = 10$ (maximum possible!)

Key Insight

At every cell $dp[i][j]$, we ask: "Is it better to include item i or not?" We do